太阳能控制器
通信协议
V1.0.0

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Version Revision History

Edition	Revision content	Expurgator	Date
V1.0.0		Leo	202603

Serial Port Parameters

Default baud rate: 115200,8-bit data bits, 1 stop bit, no parity.
Note: When using RS485 communication, a baud rate of 9600 is recommended for long lines.

Format Description

Each register is 2 bytes in size, with its address and value organized in the high-byte-first, low-byte-last format.
A 32-bit number is stored with register addresses where the high byte occupies the lower address and the low byte occupies the higher address. For example, the 32-bit data 0x12345678 is stored in registers 0x01 and 0x02, arranged in the register table as address 0x01 = 0x1234 and address 0x02 = 0x5678.

Reference Format

Device Address	FC	Data	CRC verification
1 byte	1 byte	N byte	2 byte

1) Device Address
Range: 0x01–0xF7, 0xF8–0xFF reserved address
0x00 is a broadcast address: the slave receives commands but does not return data. 0xFF is an all-purpose address: the slave receives commands and returns data, applicable only when a single slave is connected.

2) FC

0x03: Read registers (single and multiple)

3) data
Different function codes have different data structures; see the subsequent explanation for details.

4) verification
Standard CRC verification algorithm: all data from the slave address up to before the CRC check.
Note: Unlike data, the low byte of the CRC portion comes first, followed by the high byte.

The total length of a Modbus serial communication data unit is 256 bytes, with the address code, function code, and CRC check sum totaling 4 bytes, leaving up to 252 bytes for the actual data.

Read Register

Device Address	FC	Data		CRC verification
1 byte	1 byte	4 byte		2 byte
	0x03	2 Byte, starting address	2 Byte; Number of registers: 0x01–0x7d	

normal response

Device Address	FC	Data	CRC verification
1 byte	1 byte	2*N+1 bytes	2 byte

	0x03	1 Byte: Byte length of the value	2*N bytes, register value from 1 to N	

Error Return

Device Address	FC	Data	CRC verification
1 byte	1 byte	1 byte	2 byte
	0x83	See the error code table for details	

Error Code Table

Code	Name	Meaning
0x01	Unsupported function code	The device does not support this command
0x02	Illegal register address	The requested register exceeds the range
0x03	Illegal register value	The value exceeds the allowed range
0x04	Operation failed	
0x05	Length error	Frame length exceeds 256 bytes
0x06	Validation failed	CRC verification failed
0x07	Read-only parameter	
0x08	Wrong password	

Data Specification

Name	Unit	Multiplying power	Explain
Voltage	V	0.01	16 An unsigned integer ranging from 0 to 65535, corresponding to 0 V to 6553.5 V
Current	A	0.1	16 An unsigned integer with a range of 0 to 65535, corresponding to 0A to 6553.5A
Power	W	0.1	16 An unsigned integer ranging from 0 to 65535, corresponding to 0A to 6553.5W?
Quantity of electricity	Wh	0.1	
Temperature	℃	0.1	16 A signed integer with a range of-32767 to 32767, corresponding to-3276.7°C to 3276.7°C

Register Address Table

System Information Area

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x000A	1	Product ID	R			%s			
0x000B	6	Product model	R			%s			
0x0011	3	Software release	R			%s			X.X.XX
0x0014	3	Hardware Version	R			%s			X.X.XX
0x0017	3	Protocol Version	R			%s			
0x001A	6	Serial number	R			%s			
0x0020	8	Manufacturer	R			%s			
This data area address range: 0x000A–0x0027									

Reserved alternative address range: 0x0028–0x004F

product ID

ID	Name	Explain
2	LTB-24xx	BLE PWM
3	LTW-24xx	Wi-Fi+BLE PWM
4	LTM-24xx	Wi-Fi+BLE MPPT
128		Custom Development Product

System Configuration Area

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x0050	1	Maximum voltage of photovoltaic panels	R						
0x0051	1	Maximum Voltage of the Battery	R						
0x0052	1	Maximum Charging Current	R						
0x0053	1	Maximum Current Load	R						
0x0054	1	Number of Batteries	R						
0x0055	1	Configuration Item 1	R			See end of table			
0x0056	1	Configuration Item 2	R			See end of table			
0x0057	1	Configuration Item 3	R			See end of table			
This data area address range: 0x0050–0x0057 Reserved alternative address range: 0x0057–0x007F									

Configuration Item 1: B0: Supports soft power on/off; B1: Supports soft restart; B2: Supports restoring factory settings; B3: Supports firmware upgrade; B4: Supports OTA upgrade.
B8: Supports Bluetooth; B9: Supports RS232; B10: Supports RS485; B11: Supports CAN bus; B12: Supports USB; B13: Supports Wi-Fi; B14: Supports mobile networks.
Configuration Items: B0: Supports automatic voltage detection; B1: Supports USB charging port; B2: Supports USB fast charging; B3: Supports internal temperature monitoring; B4: Supports external temperature monitoring; B5: Supports temperature calibration; B6: Supports temperature compensation; B7: Supports load control; B8: Supports load short-circuit protection switch; B9: Supports load power adjustment; B10: Supports voltage calibration; B11: Supports charging current calibration; B 12: Supports discharge current calibration; B13: Supports RTC time synchronization; B14: Supports GPS functionality; B15: Supports cooling fan operation; B16: Supports cooling fan parameter adjustment.
Configuration Item 3: B0: Supports power generation statistics; B1: Supports power consumption statistics; B2: Supports USB power consumption statistics; B3: Supports battery-free applications; B4: Supports mains power supplementation; B5: Supports charging switch control; B6: Supports USB switch control; B7–15: Standby modes.

Connect Data Area

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x0080	1	GPS fixed position	R						
0x0081	2	Longitude	R						
0x0083	2	Latitude	R						
0x0085	1	Altitude	R						
0x0086	1	Bluetooth Networking Status	R			0: Not Networked			

						1: Slave Device 2: Host			
0x0087	1	Bluetooth connection	R			0: No connection; 1: Connected (Link); 2: Incorrect password; 3: Connected (App);			
0x0088	1	WIFI linkage	R			Byte0： linkage Byte 1: Signal strength			
0x0089	1	Mobile Network Connection	R			Byte0： linkage Byte 1: Signal strength			
0x008A	1	Server Connection	R			0: Disconnect; 1 Connect			
0x008B	1	Subscription Status	R			0: Failed; 1: Successful			
This data area address range: 0x0080–0x008B Reserved alternative address range: 0x008C–0x009F									

Running Data Area

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x00A0	1	Running state	R			See the end of the table for details			
0x00A1	4	Fault code	R			See the end of the table for details			
0x00A5	1	Solar panel voltage	R	0.01	V	%d			
0x00A6	1	Solar panel current	R	0.1					
0x00A7	1	Solar Panel Power	R	0.1					
0x00A8	1	Battery Type	R			Byte14-15: 00 No battery 01: Lead-acid 10: Lithium batteries 11: Reserve Bytes 8–13: ID Bytes 0–7: String Count			
0x00A9	1	Battery rated voltage	R			%d			
0x00AA	1	Accumulator voltage	R	0.01		%d			
0x00AB	1	Battery Capacity	R	0.1		%d			Percentage
0x00AC	1	Byte0: Charging Phase Byte 1: Charges switch	R			Charging stage： 0: Charge Off 1: Fast Charge 2: Equal Charge 3: Floating Charge 4: Balance 5：MPPT 6: Pause/Stop			

						Charges switch： 0: Off 1: Open			
0x00AD	1	Charging voltage	R	0.01					
0x00AE	1	Charging current	R	0.1					
0x00AF	1	Charging Power	R	0.1					
0x00B0	1	Load switch	R			0: Off, 1: On			
0x00B1	1	Load voltage	R	0.01					
0x00B2	1	Load current	R	0.1					
0x00B3	1	Bearing power	R	0.1					
0x00B4	1	USB switch	R			0: Off, 1: On			
0x00B5	1	USB voltage	R	0.01					
0x00B6	1	USB Current	R						
0x00B7	1	USB power	R						
0x00B8	1	Internal Temperature	R	0.1					
0x00B9	1	External temperature	R	0.1					
0x00BA	1	Fan Switch	R						
0x00BB	1	Fan Speed	R						
This data area address range: 0x00A0–0x00BB Reserved alternative address range: 0x00BC–0x00FF									

Operation status: 0 – Power-on delay; 1 – Upgrading in progress; 2 – Upgrade failed; 3 – Initialization; 4 – Battery activated; 5 – Running; 6 – Manual shutdown; 0 – FMP fault.

Fault Code

	Definition
ErrorCode0 Byte0	B0: Photovoltaic panels connected in reverse B1: Photovoltaic panel overvoltage protection (when exceeding the controller's maximum input voltage for the photovoltaic panel) B2~B3: Charging circuit fault: 0 – Normal; 1 – Open circuit; 2 – Short circuit. B4: Battery not connected B5: Battery reversed polarity B6: Battery overvoltage B7: Battery Overcharge Protection B8: Battery under-voltage alarm B9: Battery low-power disconnection B10: Protection for rechargeable batteries during high-temperature charging B11: Low-temperature charging protection for batteries B12: High-temperature discharge protection for batteries B13: Battery Low-Temperature Discharge Protection B14: Overcurrent protection during charging (when current exceeds 1.3 times the maximum charging current) B15: Retain
ErrorCode1	B0~B1: Load discharge circuit fault: 0 – Normal; 1 – Open circuit; 2 – Short circuit.

Byte1	B2–B3: Load protection status: 0 – Normal; 1–1.1 – Overcurrent; 2–1.3 – Overcurrent; 3 – Load off; B4–B5: USB fault: 0 – Normal; 1 – Overvoltage; 2 – Low voltage; 3 – Overcurrent; B6~B7: 0 – Normal; 1 – Internal high temperature; 2 – High-temperature current limiting; 3 – High-temperature protection;
ErrorCode2 Byte2	B0: GPS cannot determine location; B1: wrong password ; B2: WIFI connection lost; B3: Mobile network connection lost; B4: Server not connected; B5: Server warning flag; B6: Time synchronization failed; B7: Keep
ErrorCode3 Byte3	Keep

Event Log

Controller Event

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x0100	1	Total Number of Logs	R						
0x0101 ~ 20*6	6	Log data: one record uses 6 registers. Word 0: Event Code Word 1: Battery Voltage Words 2–3: Date Character 4–5: Hour, Minute, Second	R						
This data area address range: 0x0100–0x0177 Reserved alternative address range: 0x0178–0x01FF									

Communication Event

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x0200	1	Total Number of Logs	R						
0x0201 ~ 20*6	6	Log data: one record uses 6 registers. Word 0: Event Code Word 1: Battery Voltage Words 2–3: Date Character 4–5: Hour, Minute, Second	R						
This data area address range: 0x0200–0x0277 Reserved alternative address range: 0x0278–0x02FF									

Event Coding

0	Invalid Data
1	Power on the device (initialization complete)
2	Device powered on (initialization failed). Note: Communication module connection failed.
3	Start Charging
4	End Charging
5	Enable Load
6	Unload
7	The charging circuit is open.
8	Short circuit in the charging circuit
9	Load circuit open
10	Load circuit short circuit
11	Battery low charge alarm
12	Battery low-voltage protection
13	Battery overpressure
14	Internal Overheat Protection
15	Load short-circuit protection
16	Enable charging
17	Turn off charging
18	Enable USB Output
19	Turn off USB Output
20	Photovoltaic panel overvoltage protection
21	Overcurrent Protection During Charging
22	The battery is not connected.
20···127	······
128	Incorrect Bluetooth connection password
129	Bluetooth connection successful
130	WIFI connection failed: Connection timed out or busy, no response
131	WIFI connection successful
132	MQTT connection failed: Connection timed out or busy with no response
133	MQTT connection successful
134	MQTT subscription failed: Connection timed out or busy with no response
135	MQTT subscription successful
136	······

Statistics Area

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x0300	2	Total Running Time	R		Second				
0x0302	2	Cumulative Power Generation	R	0.1	KWh				
0x0304	2	Total Electricity Consumption	R	0.1	KWh				
0x0306	1	Total number of full charges for the battery	R						

0x0307	1	Total number of over-discharge cycles for the battery	R						
0x0308	1	Electricity generated on that day	R	0.1	Wh				
0x0309	1	The maximum voltage of the photovoltaic panels on that day	R	0.01	V				
0x030A	1	The maximum current of the photovoltaic panels on that day	R	0.1	A				
0x030B	1	The maximum power of the photovoltaic panels on that day	R	0.1	W				
0x030C	1	The maximum voltage of the battery on that day	R	0.01	V				
0x030D	1	The lowest battery voltage on that day	R	0.01	V				
0x030E	1	Electricity consumption on that day	R	0.1	Wh				
0x030F	1	Maximum Current at Load Limit on That Day	R	0.1	A				
0x0310	1	Maximum power load on that day	R	0.1	W				
0x0311	1	USB power consumption on that day	R	0.1	Wh				
0x0312	1	Maximum USB current on that day	R	0.1	A				
0x0313	1	Maximum USB power on that day	R	0.1	W				
0x0314	1	Electricity generated yesterday	R	0.1	Wh				
……	1	The maximum voltage of the photovoltaic panels yesterday	R	0.01	V				
This data area address range: 0x0300–0x0378 Reserved alternative address range: 0x0379–0x04FF									

Statistical Log

Address	Length	Name	RW	Multiplying power	Unit	Form	Windows default	Value Range	Explain
0x0300 ~ 7*16	1	Log data: one record uses 12 registers. See the daily data in the table above, pages 13–14: Date Words 15–16: Hours, Minutes, Seconds	R						